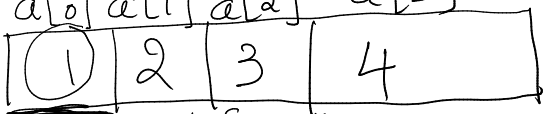
int a[] = {1, 2, 3, 4} ⇒ element repn



1000 1004 1008 1012
a a+1 a+2 a+3 ⇒ address repn

a[0] = *(a+0)
= *(1000+0) ⇒ *(1000) ⇒ 1

a[2] = *(a+2)
= *(1000 + 2×4) ⇒ 1008 ⇒ 3

fun(a) ⇒ 0th elem
↳ address
fun(int (*p))

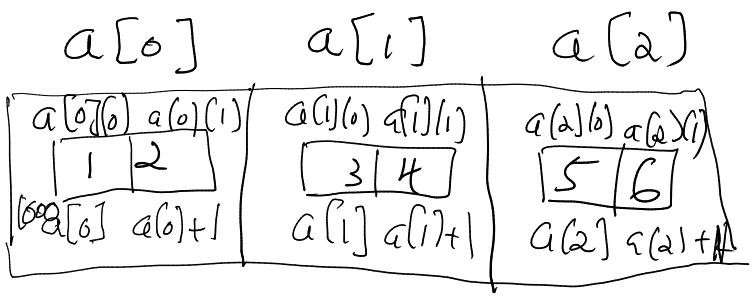
2D-array

↳ collection of 1D array, each 1D array is a collection of elements.

datatype arrayname [rows] [cols];

int arr [2] [3]
row, col

3 → 1D array
each 1D ⇒ 2 elements.



arr[3][2]

a[0] = *(a+0)

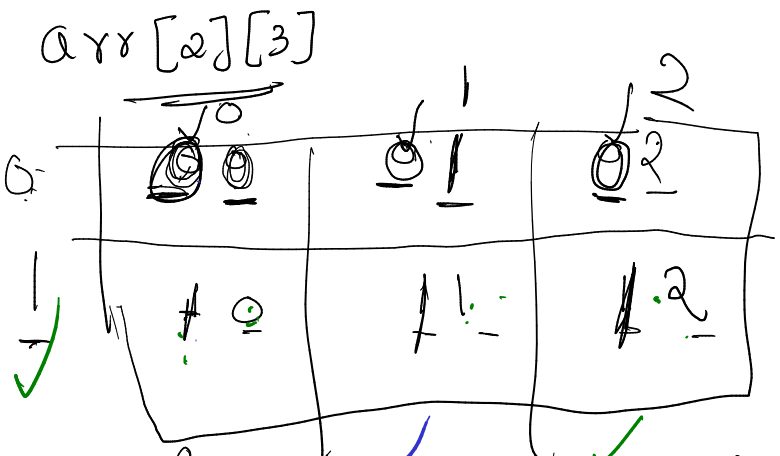
a 1000 a a+1 a+2

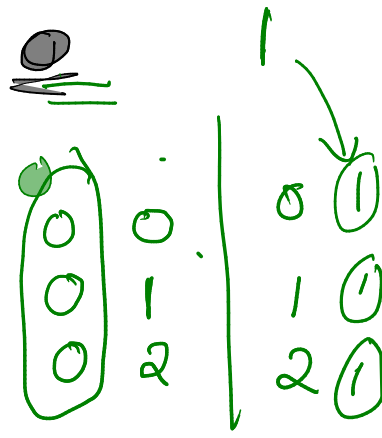
a[0][0] ⇒ * (*(a+0) + 0)

* (*(a+1) + 0)

zeroth row zeroth col

int (*ptr)[2]





for (i=0; i<2; i++)

0<2

i=0, 0<2 j=0

a[0][0]

j=1, 1<3

a[0][1]

j=2, 2<3 a[0][2]

{

for (j=0; j<3; j++)

{

scanf("%d", &a[i][j]);

}

}

$a = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ 2×3

$b = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ 2×3

$c[2][3] = \begin{bmatrix} 1+1 & 2+1 & 3+1 \\ 1+1 & 1+1 & 2+1 \end{bmatrix}$

$c[2][3] = \begin{bmatrix} 2 & 3 & 4 \\ 2 & 2 & 3 \end{bmatrix}$

a[2][3]

	0	1	2
0	00 1	01 2	02 3
1	10 1	11 1	12 2

b[2][3]

b	0	1	2
0	00 1	01 1	02 1
1	10 1	11 1	12 1

i=0

a[0][0] + b[0][0]
a[0][1] + b[0][1]
a[0][2] + b[0][2]

i=1

a[1][0] + b[][0]
j=0
a[i+1][1] + b[1][1]
a[1][2] + b[1][2]

c[2][3]

$c[2][3] = \begin{bmatrix} 00 & 01 & 02 \\ 1 & 3 & 4 \\ 10 & 11 & 12 \\ 2 & 2 & 3 \end{bmatrix}$

```
for (i=0; i<2; i++)
{
  for (j=0; j<3; j++)
  {
    c[i][j] = a[i][j] + b[i][j];
  }
}
```

a[2][3] 2 row, 3rd column

	0	1	2
0	1	2	3
1	4	5	6

2x3

	0	1	2
0	00 1	01 2	02 3
1	10 4	11 5	12 6

1 => 00
2 => 01
3 => 02

4 => 10 ✓
5 => 11
6 => 12

	0	1
0	00 1	01 10
1	10	11
2	01	
	20 02	

Transpose => 3x2

T[3][2] = 0

3 row, 2 col

	0	1
0	00 1	01 4
1	10 2	11 5
2	20 3	21 6

```
for(i=0; i<2; i++)
{
    for(j=0; j<3; j++)
    {
        t[j][i] = arr[i][j];
    }
}
```

t[3][2]

t[0][0] = a[0][0]
t[0][1] = a[1][0]
t[1][0] = a[0][1]
t[1][1] = a[1][1]

i=0, 0<2
j=0, 0<3

t[0][

3

	0	1
0	1 1	4 0 1
1	2 1 0	5 1 1
2	3 2 0	6 2 1

t

```
for(i=0; i<3; i++)
    for(j=0; j<2; j++)
    {
        t[i][j] =
    }
```